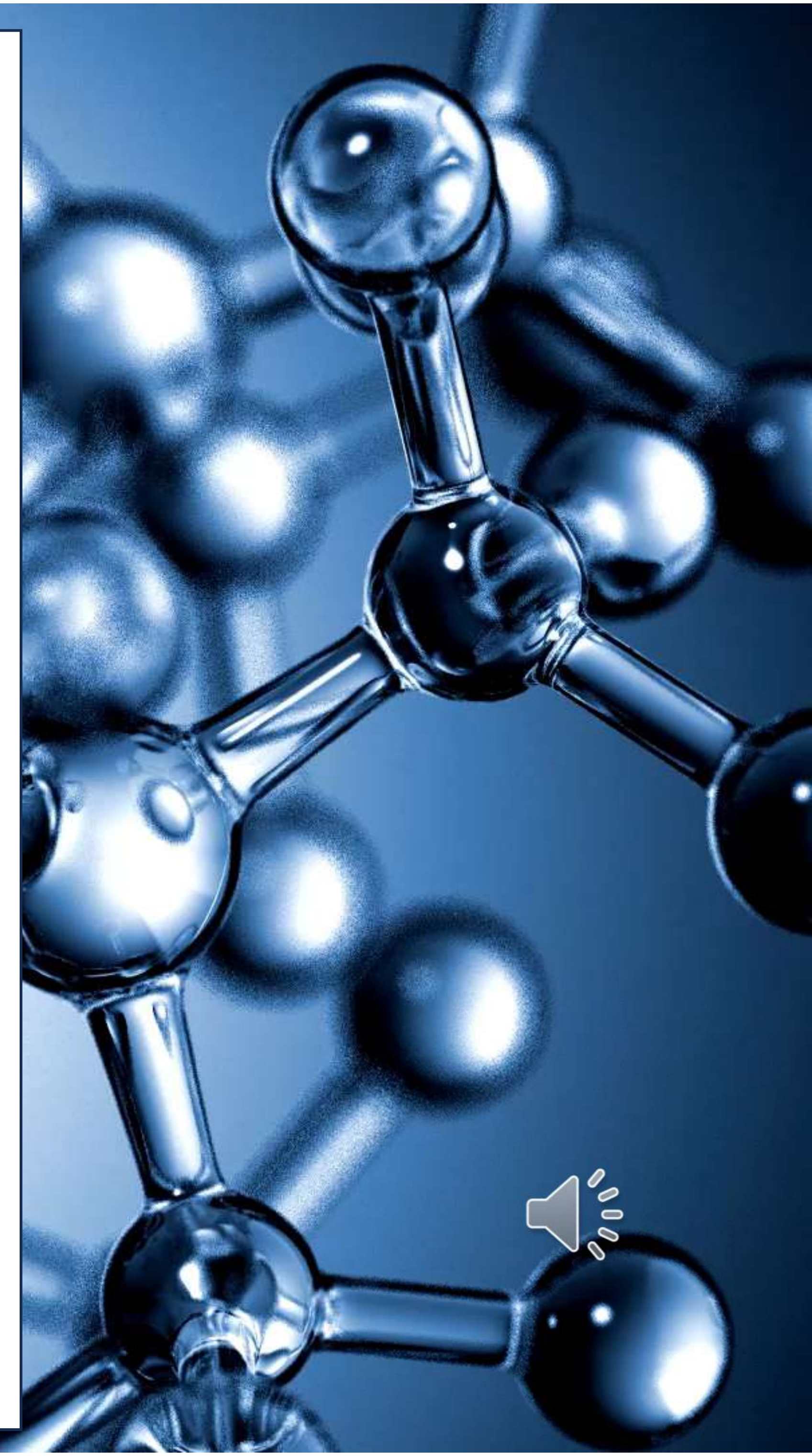
WELCOME!

ISO 13485:2016

Medical Devices
Quality management systems
Requirements for regulatory purposes



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YOUR INSTRUCTOR

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BEng, MEng (Electronics and Software Engineering).
Over 20 years working experience in highly regulated industries.
Working in the medical devices industry since 2011.
Expertise in ISO 13485 requirements and implementation for Quality and Regulatory compliance, QMS auditing, product development, manufacturing and test process automation, validation, servicing and sustaining engineering activities.

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COURSE OBJETIVES

1) Review the purpose of a Quality Management System (QMS).
2) Understand the scope and requirements of ISO 13485:2016
3) Learn how to implement those requirements in your organization.
4) Identify which actions are required to ensure the continuous compliance with the standard.

After completing this course, you will have the **tools** and **knowledge** to support the implementation and maintenance of a QMS, in compliance with **ISO 13485:2016**

COURSE STRUCTURE AND MATERIAL

- 8 Lessons.
- 6 Assignments throughout the course.
- 1 Final Assignment.

and...

- Online access to all lessons.
- Copy of PowerPoint slides and assignments in PDF.
- Copy of list of references and bibliography in PDF.
- Certificate of completion.

COURSE CURRICULUM

- **Lesson 1:** Why do we need a Quality Management System?
- **Lesson 2:** Scope and applicability of ISO 13485:2016
- Assignment 1
- **Lesson 3:** General QMS requirements
- **Lesson 4:** Documentation requirements
- Assignment 2

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COURSE CURRICULUM

- **Lesson 5:** Management responsibility
- **Lesson 6:** Resource management
- Assignment 3
- **Lesson 7:** Product realization
- Assignments 4 and 5

COURSE CURRICULUM

- **Lesson 8:** Measurement, analysis and improvement
- Assignment 6
- Final Assignment

LESSON 1: Why do we need a QMS?

- A Quality Management System is a set of tools and methods used to assure the quality, effectiveness and safety of a company's products or services.
- It provides a systematic approach to describe and control any critical activity or operation used by the organization.
- Using a QMS can make the organization be more competitive, reduce its running costs, increase revenue and improve customer's satisfaction.
- Some countries require the implementation of a QMS, to allow companies to distribute their products or services.

LESSON 1: Why do we need a QMS?

- Implementing a common industry standard (like ISO 13485) in the QMS enables different stakeholders to interact more efficiently (same rules).
- ISO 13485 follows a risk-based approach, focused on meeting customer, safety and regulatory requirements.
- Compliance with ISO 13485 is the baseline which allows companies in the medical devices industry to sell their products and services worldwide.
- Additional country-specific regulations may apply, depending on the type of products and services provided.

LESSON 2: Scope and applicability

- ISO 13485 covers all the phases and activities of the product life-cycle.
- It applies to organizations that design, produce, store or distribute, install, service or repair medical devices, directly or as a subcontractor.
- The organization may not need to meet all requirements of ISO 13485.
- The organization must describe how the applicable clauses of the standard are fulfilled, and why some requirements are not applicable.
- Any exception must be clearly defined and justified.

LESSON 2: Scope and applicability

- For the applicable requirements, the organization must:

1) Identify the relevant customer and regulatory requirements.
2) Document all relevant requirements as QMS policies and procedures.
3) Establish a risk-based approach to the applicable processes.
4) Implement, control, maintain and improve all applicable documents, operations, processes and resources for continuous regulatory compliance.

A Process is a set of activities that convert inputs into outputs, to accomplish a specific goal.

ASSIGNMENT 1

- **Lesson 1:** Why do we need a Quality Management System?
- **Lesson 2:** Scope and applicability of ISO 13485:2016

LESSON 3: General QMS requirements

- The Quality Management System shall be documented and maintained.
- The QMS must be effective at meeting the requirements of the ISO 13485 standard and other applicable regulatory requirements.
- Any relevant requirement, process, activity, arrangement, or procedure shall be identified, included, planned, implemented, and controlled in the QMS.

The overall document that describes how the organization is implementing the relevant requirements of the ISO 13485 standard is called the Quality Manual.

LESSON 3: General QMS requirements

- The roles undertaken by the organization (manufacturer, authorized representative, importer, or distributor) shall be documented.
- The roles of the organization will determine which activities and controls the organization must implement in its QMS.
- The organization shall determine which processes are needed for the quality management system and how they must be applied, based on the roles undertaken by the organization.

LESSON 3: General QMS requirements

- The organization shall apply a risk-based approach to the control of the required processes.
- The sequence and interaction of the processes must be determined.
- The criteria and methods for ensuring that both the operation and control of processes of the QMS are effective must be determined.
- Resources and information necessary to support the operation and monitoring of the QMS must be available.

LESSON 3: General QMS requirements

- The organization shall define actions necessary to achieve planned results and maintain the effectiveness of the QMS processes.
- The QMS processes shall be monitored, measured as appropriate, and analyzed.
- Records of the QMS processes, needed to demonstrate compliance to ISO 13485 and other applicable regulatory requirements, shall be established and maintained.

Records are sometimes referred to as objective evidence.

If it is not documented, it didn't happen!

LESSON 3: General QMS requirements

- Changes to any process shall be controlled and evaluated.
- Evaluation must be based on the potential risk of those changes to the quality system and the product or services provided by the organization.
- The organization may choose to outsource some processes.
- If the outsourced processes can impact the conformity of the products or services, those processes shall be monitored and controlled.

LESSON 3: General QMS requirements

- The organization remains responsible of meeting customer, safety and regulatory requirements for those outsourced processes.
- The controls over outsourced processes shall be proportionate to the risk involved and the ability of the external party to meet the requirements.
- The controls shall include written Quality Agreements.

A Quality Agreement is a written contract between the organization and the external party, to define the quality – related activities and ownership.

LESSON 3: General QMS requirements

- Computer software applications used to support the QMS activities must be validated prior to initial use, following a validation procedure.
- Changes to the software or its intended use may require revalidation.
- The extend of the validation effort must be based on risk.
- Records of the validation must be maintained.

Validations are usually completed in 3 sequential steps, i.e., IQ (Installation Qualification), OQ (Operational Qualification) and PQ (Performance Qualification).

LESSON 4: Documentation requirements

- The Quality Management System documentation shall include:

1) A documented statement of a Quality Policy and Quality Objectives.
2) A Quality Manual.
3) Procedures and records required by ISO 13485, as applicable.
4) Additional documents and records determined by the organization, for the effective planning, operation, and control of processes, and as required by other applicable regulatory requirements.

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LESSON 4: Documentation requirements

- The Quality Manual shall include:

1) The scope of the quality management system, including a list of the exclusions and nonapplications, and their justification.
2) The list of relevant QMS procedures, or a reference to them.
3) A description of the interactions between the processes of the quality management system.
4) A description of the structure of the documentation system used by the quality management system.

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LESSON 4: Documentation requirements

- The organization must develop a Medical Device File, specific to each product model, type, class, or category.
- The file must include the relevant documents describing the medical device specifications, and realization, servicing and installation processes.
- The content of this file shall demonstrate compliance to the requirements of the ISO 13485 standard and other applicable regulatory requirements.

LESSON 4: Documentation requirements

- The minimum contents of the Medical Device File are:

1) General description of the medical device, intended use or purpose, and labelling, including instructions for use.
2) Specifications for product.
3) Specifications or procedures for manufacturing, packaging, storage, handling, and distribution.
4) Procedures for measuring and monitoring.
5) Requirements for installation and servicing (if applicable).

LESSON 4: Documentation requirements

- The organization shall control the documentation and records necessary for the operation of the quality management system.
- A procedure shall be created to describe the controls applicable to:

  a) Review and approval of both new and updated documents, prior to release for use.

  b) Identification of documentation revision status and change history.

LESSON 4: Documentation requirements

c) Ensure the controlled distribution of documents, so only the relevant versions are available for use.

d) Ensure the documents remain legible and identifiable.

e) Prevent deterioration and loss of documents.

f) Identify and prevent the unintentional use of obsolete documents.

Identification and distribution requirements also apply to documents generated outside the organization, that have been identified as necessary for the operation of the QMS.

LESSON 4: Documentation requirements

- Changes to documents must be reviewed and approved by personnel with suitable background information, to support their decisions.
- The retention period of a copy of obsoleted documents must be defined.
- Documents used in the manufacture and test process of medical devices must be retained for the lifetime of the product, or as specified by applicable regulatory requirements.
- Records shall be generated and maintained, as evidence of conformity to applicable requirements.

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LESSON 4: Documentation requirements

- A procedure must be created to describe the control of records.
- The control methods shall ensure the records are identified, stored, secured, retrievable, retained and disposed.
- Confidential information contained in records must be protected, according to applicable regulatory requirements.
- Records must remain legible, identifiable and retrievable.
- Changes to records must remain identifiable.

Changes to records must follow Good Documentation Practices (GDP).

LESSON 4: Documentation requirements

- Records shall be retained for at least the lifetime of their relevant medical device, according to the manufacturer's specifications, but not less than 2 years since the medical device was released to the market.
- If applicable regulatory requirements demand other retention periods, they must be also applied.

ASSIGNMENT 2

- **Lesson 3:** General QMS requirements
- **Lesson 4:** Documentation requirements

LESSON 5: Management responsibility

- The top management team must provide evidence of its commitment to the development, implementation and effectiveness of the QMS, as follows:

1) Communicating to the organization the importance of meeting customers' and regulatory requirements.

2) Documenting the Quality Policy and Quality Objectives.

3) Conducting Management Reviews, to verify the effectiveness of the QMS.

4) Ensuring the availability of resources.

LESSON 5: Management responsibility

- Top management must ensure that customer and regulatory requirements are determined and met.
- Top management must ensure that the Quality Policy:
  a) Applies to the purpose of the organization.
  b) States the organization's commitment to comply with the applicable requirements and to maintain the effectiveness of the QMS.
  c) Provides a framework for setting and reviewing quality objectives.

LESSON 5: Management responsibility

- Top management must ensure that the Quality Policy is communicated and understood within the organization.
- The Quality Policy must be periodically reviewed and updated if necessary.
- Top management must ensure that Quality Objectives are:
  a) Established to meet the relevant regulatory and customer requirements.
  b) Allocated to the appropriate functions and processes.
- The Quality Objectives shall be measurable and consistent with the Quality Policy.

LESSON 5: Management responsibility

LESSON 5: Management responsibility

- Top management must ensure that:

1) The planning of the QMS is carried out, to meet the applicable requirements and Quality Objectives.

2) The integrity of the quality management system is not impacted when changes are planned and implemented.

LESSON 5: Management responsibility

3) Authorities and responsibilities are defined, documented and communicated within the organization.

4) The interrelations of all personnel who manage, perform, and verify work affecting quality is documented.

That personnel shall have the independence and authority necessary to perform those tasks effectively.

LESSON 5: Management responsibility

- Top management shall appoint a management representative, with the following authority and responsibility:

1) Ensure the documentation and implementation of processes necessary for the operation of the QMS.

2) Report to the top management on the effectiveness of the QMS, and on opportunities for improvements.

3) Promote awareness of applicable regulatory and quality requirements throughout the organization.

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LESSON 5: Management responsibility

- Top management shall ensure that appropriate processes are implemented within the organization, to communicate the effectiveness of the QMS.
- A procedure must be created to describe the Management Review process, to be completed at defined intervals by top management, to:
  a) Review the QMS to ensure its continuous suitability and effectiveness.
  b) Assess opportunities for improvement of the QMS.
  c) Assess the need for changes to the Quality Policy and Quality Objectives.

LESSON 5: Management responsibility

- The input to the Management Review process must include:

1) Results of feedback activities.

2) Handling of complaints from customers and other stakeholders.

3) Reporting to regulatory authorities.

4) Audit results.

5) Results of monitoring and measurement activities of products and processes.

6) Corrective and Preventive Actions (CAPA) process.

LESSON 5: Management responsibility

7) Status of follow-up actions agreed in previous reviews.
8) Changes that could affect the QMS, like new or revised regulatory requirements.
9) Recommendations for improvements.
10) Other inputs, as defined by the organization.

LESSON 5: Management responsibility

- The output of the Management Review process shall be recorded and maintained. That output must include the inputs reviewed, the decisions made and the agreed actions, regarding:
  a) Improvements to the QMS, to maintain its suitability, adequacy, and effectiveness, or changes due to new or revised regulatory requirements.
  b) Improvements or changes to products and services, related to customer's requirements.
  c) Resources needed to implement those changes.

LESSON 6: Resource management

- The organization shall determine and arrange resources required to:
  a) Implement the QMS and maintain its effectiveness.
  b) Meet the customer and regulatory requirements.
- Personnel performing any activity that may affect product quality must be competent in terms of education, training, skills, and experience.
- The process for establishing competence, training needs, personnel awareness, must be documented.

LESSON 6: Resource management

- The organization shall ensure that:

1) The necessary competence level (education, experience) is determined, for the personnel performing work that affects product quality.

2) Training is provided, for the relevant personnel to achieve and maintain the necessary competence.

3) The effectiveness of training activities is evaluated.

4) The personnel is aware of the importance of their activities, to achieve the Quality Objectives.

LESSON 6: Resource management

5) Records are maintained as evidence of the personnel's education, training, skills and experience.

The methodology for evaluating the effectiveness of training activities shall be proportionate to the risk associated with the work for which the training is being provided.

LESSON 6: Resource management

- The organization shall document the requirements for the infrastructure required to:

1) Achieve conformity to product requirements.

2) Prevent product mix-up and ensure orderly handling of product.

- Infrastructure includes buildings, workspaces, utilities, process equipment (hardware and software) and supporting services (e.g., transport, communication, or information systems).

LESSON 6: Resource management

- The required infrastructure maintenance activities and intervals shall be defined and documented, if they can affect the product quality.
- Maintenance activities may include equipment used in production, the control of the work environment, and monitoring and measurement.
- Records of the maintenance activities shall be maintained as evidence.

The periodic calibration and inspection of monitoring and measurement equipment is a typical maintenance activity in manufacturing environments.

LESSON 6: Resource management

- The organization shall:

1) Document the requirements for the work environment, required to achieve conformity to product requirements, and the procedures to monitor and control the specified conditions of the work environment.

2) Document the requirements for the control of personnel health, cleanliness and clothing, if contact between such personnel and the product or work environment could affect the product quality.

LESSON 6: Resource management

3) Ensure that personnel temporary working under special environmental conditions within the work environment are competent to complete the required tasks, or are supervised by a competent person.

Refer to standards ISO 14644 and ISO 14698 for additional information regarding Cleanrooms and associated controlled environments (Classification of air cleanliness by particle concentration and Biocontamination control).

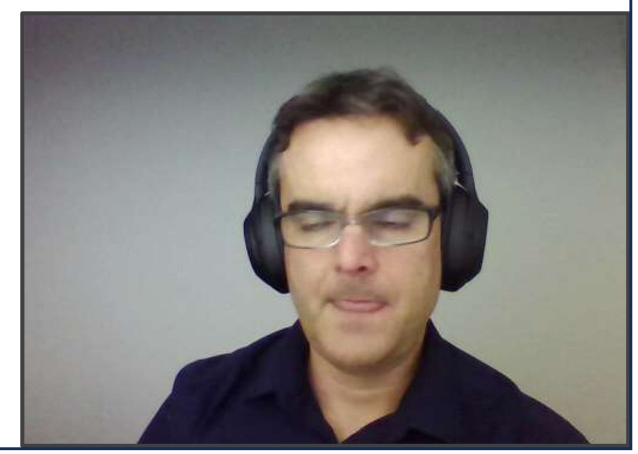
LESSON 6: Resource management

4) Plan and document the actions to control contaminated or potentially contaminated products, to prevent the contamination of the work environment, personnel, or other products.

5) For sterile products, the requirements for controlling contamination with microorganisms or particulate matter must be documented. The required cleanliness during assembly or packaging processes must be maintained.

ASSIGNMENT 3

- **Lesson 5:** Management responsibility
- **Lesson 6:** Resource management



LESSON 7: Product realization

- The organization shall:

1) Plan and develop the processes needed for the realization of the product.
2) Planning activities shall be consistent with the requirements of the other processes of the Quality Management System.
3) Apply a Risk Management process as part of product realization, and maintain the records generated as part of those activities (e.g., pFMEA).

A Process Failure Mode and Effects Analysis (pFMEA) is a method used to identify and mitigate risks related to a specific process or set of activities.

LESSON 7: Product realization

- As part of product realization planning, the organization shall:

1) Define Quality Objectives and requirements for the product.
2) Identify the processes, documents and resources (including infrastructure and work environment) required for the product.
3) Establish the required verification, validation, monitoring, measurement, inspection and test, material handling and storage, distribution and traceability activities to be used, and the criteria to be applied for the acceptability of product.

LESSON 7: Product realization

4) Define which records must be generated as evidence of the realization of product meeting all applicable customer and regulatory requirements.

- The output of the product realization planning activities shall be documented in a suitable form.

Refer to standard ISO 14971 for additional information regarding the application of risk management to medical devices.

LESSON 7: Product realization

- The organization shall determine:

1) The product requirements as defined by the customer, including delivery and post-delivery activities.
2) Additional product requirements that may not explicitly defined by the customer, but are needed based on the intended use of the product.
3) The regulatory requirements applicable to the product.
4) The user training required to ensure the safe and effective use of the product.

LESSON 7: Product realization

- All product requirements identified must be then reviewed by the organization, prior to committing to supply those products to the customer.
- That review shall ensure that:
  a) All product requirements are defined and documented.
  b) Discrepancies on customer requirements are resolved.
  c) Regulatory requirements are met.
  d) User training can be provided.
  e) The organization can meet those requirements.

LESSON 7: Product realization

- Records of the product requirements review, and actions list must be maintained.
- If the customer doesn't provide a list of requirements, the list must be documented and confirmed by the organization.
- The relevant documents must be updated if the product requirements change, and the relevant personnel must be informed.

LESSON 7: Product realization

- Communication with customers shall be planned and arranged following a documented process, when sharing information like:
  a) Product information.
  b) Enquiries, contracts and product ordering.
  c) Customer feedback and complaints handling.
  d) Advisory notices.
- Communication with regulatory authorities shall follow applicable regulatory requirements.

LESSON 7: Product realization

- Procedures for the product Design and Development process shall be documented, to plan and control those activities.
- As part of Design and Development planning, the organization shall document the following:

1) The design and development phases.
2) The reviews required at each phase.
3) The design verification, validation and transfer to manufacturing activities, required at each process stage.

LESSON 7: Product realization

4) The responsibilities and authorities for design and development actions.
5) The methods to be used to ensure traceability between design and development inputs and outputs.
6) The resources and competence of the personnel needed to complete the product design and development activities.

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LESSON 7: Product realization

- The inputs to the Design and Development process must be documented and include all relevant product requirements, such as:

1) Functional, performance, usability, and safety requirements, according to the intended use of the product.
2) Applicable regulatory requirements and standards.
3) Applicable outputs of Risk Management activities (e.g., dFMEA, aFMEA).
4) Information derived from other similar products, as applicable.

LESSON 7: Product realization

5) Other requirements essential for design and development of the product and processes.

- The input requirements to the Design and Development process shall be:
  a) Complete and unambiguous.
  b) Able to be tested (i.e., verified or validated).
  c) Not in conflict with each other.
- The organization shall review the inputs for adequacy and approve them.

LESSON 7: Product realization

- The outputs to the Design and Development process must be documented, approved, and verifiable against the inputs, to confirm those outputs:

1) Meet the design and development input requirements.
2) Provide sufficient information for purchasing, production and servicing.
3) Contain the product's acceptance criteria.
4) Specify the product characteristics to ensure its safe and proper use.

LESSON 7: Product realization

- At suitable stages of the Design and Development process, reviews shall be performed to evaluate the ability of the results of design and development to meet requirements and to identify and propose necessary actions.
- Participants of the reviews shall include representatives relevant to the design and development stage, and other specialist personnel.
- Records of the review shall be maintained, including identification of the reviewed design, results, actions, participants and date.

LESSON 7: Product realization

- Verification of the Design and Development outputs shall be executed, to confirm they met the input requirements.
- Design Verification plans must be developed and include methods, acceptance criteria and statistical techniques including sample size rationale (as required).
- If the intended use of the product includes the interface with other devices, verification activities must include confirmation of the correct implementation of that functionality.

LESSON 7: Product realization

- Records of the Design and Development verification activities shall be maintained, including the results and conclusions obtained.
- Validation of the resulting product shall be executed, to confirm it is capable of meeting the specified intended use.
- Design Validation plans must be developed and include methods, acceptance criteria and statistical techniques including sample size rationale (as required).

LESSON 7: Product realization

- Design Validation is executed using samples of representative products, based on initial production units, or equivalent.
- The rationale for the product used for Design Validation shall be recorded.
- Clinical or performance evaluations shall be performed as part of Design Validation, as required by regulatory requirements.

Providing a medical device to a customer or clinician, only for the purposes of evaluation (under validation) is not considered as a product release for use.

LESSON 7: Product realization

- If the intended use of the product includes the interface with other devices, validation must include confirmation of the correct implementation of that functionality.
- All validation activities shall be completed prior to releasing the product for use by the customer.
- Records of the Design and Development validation activities shall be maintained, including the results and conclusions obtained.

LESSON 7: Product realization

- The organization shall maintain documented procedures for transfer activities of design and development outputs to manufacturing.
- These procedures shall ensure that design and development outputs are verified as suitable for manufacturing, before becoming final production specifications, and that production capability can meet product requirements.
- The results and conclusions of the transfer activities shall be documented and maintained.

LESSON 7: Product realization

- The organization shall maintain documented procedures to control design and development changes.
- The significance of the change to the product function, performance, usability, safety, and applicable regulatory requirements shall be determined.
- Design and development changes shall be identified, reviewed, verified, validated (as applicable) and approved.

LESSON 7: Product realization

- The design change review shall be evaluated, to determine:
  a) How the changes will affect constituent parts, products being currently produced or already delivered to the market.
  b) Inputs or outputs of risk management files (e.g., pFMEA, dFMEA, aFMEA).
  c) Product realization processes.
- Records of design changes, their review and actions performed must be maintained.

LESSON 7: Product realization

- The organization shall maintain a Design and Development File to store the relevant records and documentation, related to the design and development of each medical device type or family.
- The file shall include references to the relevant records and other objective evidence, to demonstrate continuous product conformity against the design requirements, throughout the product lifecycle (including design changes).

ASSIGNMENT 4

- **Lesson 7:** Product realization

LESSON 7: Product realization

- The organization shall maintain documented procedures to ensure that purchased product (material, components, etc.) conforms to specifications.
- The criteria for evaluation and selection of suppliers shall be established.
- That criteria shall be proportionate to the risk associated with the medical device, and shall be based on:

a) The supplier's ability to provide a product that meets the organization's requirements, and the performance of the supplier.

c) The effect of the purchases on the product quality.

LESSON 7: Product realization

- The organization shall plan the monitoring and re-evaluation of suppliers, taking in to account the supplier's performance in meeting requirements for the purchased product.
- The results of the monitoring activities are inputs for the re-evaluation.
- Nonfulfillment of purchasing requirements shall be addressed with the supplier, based on risk and applicable regulatory requirements.
- Records shall be maintained for supplier's evaluation, selection, monitoring, re-evaluation, and related actions.

LESSON 7: Product realization

- Purchasing information shall describe the purchased product, including:
  a) Specifications and requirements for product acceptance, processes, procedures, and the use of tools and equipment.
  b) Qualification requirements for supplier personnel relevant to the purchased product.
  c) Quality management system requirements.
- The organization shall ensure the adequacy of the specified purchasing requirements, prior to placing orders to the supplier.

LESSON 7: Product realization

- A written agreement shall be included in the purchasing information, for the supplier to notify the organization of changes in the purchased product, as applicable, if those changes can affect the ability of the purchased product to meet specified purchase requirements.
- Purchasing information and records shall be maintained, for traceability.
- The organization shall ensure that purchased product meets the specified requirements, by implementing inspection and other verification activities.

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LESSON 7: Product realization

- The extent of verification activities shall be based on the supplier's evaluation results and the risks associated with the purchased product.
- In the event of changes to the purchased product, the organization shall determine how they affect the product or realization process.
- If purchased product verification is conducted by the supplier, the purchasing information shall include the inspection method, activities and acceptance criteria.
- Records of verification shall be maintained.

LESSON 7: Product realization

- To ensure product conformity to specifications, the production and service provision shall be planned, carried out, monitored and controlled.
- As appropriate, production controls shall include:

  1) Documentation of procedures and methods.

  2) Qualification of infrastructure.

  3) Implementation of monitoring and measurement of process parameters and product characteristics, including required equipment.

LESSON 7: Product realization

4) Implementation of defined operations for labeling and packaging.

5) Implementation of product release, delivery, and postdelivery activities.

- The organization shall establish and maintain a record for each medical device or batch of medical devices, to provide traceability and to identify the amount manufactured and amount approved for distribution.
- That record shall be verified and approved.

LESSON 7: Product realization

- The organization shall document cleanliness and contamination control requirements of product, if the product:
  a) is cleaned by the organization prior to sterilization or its use.
  b) is supplied nonsterile and is to be subjected to a cleaning process prior to sterilization or its use.
  c) cannot be cleaned prior to sterilization or its use, and its cleanliness is of significance in use.

LESSON 7: Product realization

d) is supplied to be used nonsterile, and the product cleanliness is important during use.

e) process agents are to be removed from product during manufacture.

When the product is cleaned in accordance with cases a) and b) above, the work environment requirements described earlier are not applicable, prior to the product cleaning process.

LESSON 7: Product realization

- The organization shall document requirements for the installation of the medical device, including acceptance criteria for verification of installation, as appropriate.
- If the medical device installation will be performed by an external party, the organization shall provide documented requirements for the installation and verification of installation.
- Records of medical device installation shall be maintained.

LESSON 7: Product realization

- If servicing of the medical device is a specified requirement, the organization shall:

1) Document servicing procedures, materials and measurements, required to complete those activities and verify that product requirements are met.

2) Analyze records of servicing activities carried out by the organization or its supplier, to identify potential complaints, and as input to the QMS improvement process.

3) Maintain records of servicing activities.

LESSON 7: Product realization

- Records of the process parameters used for each sterilization batch shall be maintained, and shall be traceable to each production batch.
- The organization shall validate any production and service provision processes, where the resulting output cannot be verified by subsequent monitoring or measurement.
- Validation shall establish the ability of those processes to achieve the expected results consistently.

LESSON 7: Product realization

- Procedures to perform process validations shall be documented, including:
  a) Defined criteria for review and approval of the processes.
  b) Qualification of equipment and personnel.
  c) Use of methods, procedures, and acceptance criteria.
  d) Statistical techniques with rationale for sample sizes, as applicable.
  e) Requirements for validation records.
  f) Revalidation and criteria for revalidation.
  g) Approval of changes to the processes.

LESSON 7: Product realization

- Procedures shall be documented by the organization, for the validation of the application of computer software used in production and service provision (also when used in the monitoring and measurement of product, to determine its conformity to determined requirements).
- Software applications shall be validated prior to initial use, and after changes to such software or its intended use, as applicable.
- The effort associated with software validation shall be proportionate to the risk of the intended use of the software, and the potential impact to product quality.

LESSON 7: Product realization

- Records of the results, conclusion and necessary actions derived from the validation shall be maintained.
- The organization shall document procedures for the validation of processes for sterilization and for sterile barrier systems, and for their revalidation following product or process changes, as appropriate.
- Relevant documentation and records shall be maintained.

Refer to standards ISO 11607-1 and 11607-2 for additional information regarding sterilized packaging validation.

LESSON 7: Product realization

- The organization shall establish procedures for product identification.
- Products shall be identified by suitable means throughout product realization.
- The status of products, regarding their conformity to monitoring and measurement requirements, shall be identified and maintained throughout product realization, including production, storage, installation, and service and repair activities.

LESSON 7: Product realization

- The status identification shall ensure that only products that have passed the required inspections and tests, or are released under an authorized concession, are dispatched, used, or installed.
- A system shall be implemented to assign a unique device identification (UDI) to the medical device, if applicable for regulatory compliance.
- Procedures shall be established to ensure that products returned to the organization are identified and distinguished from conforming products.

LESSON 7: Product realization

- The organization shall establish and maintain documented procedures for product traceability, according to the applicable regulatory requirements, and define the records required to support traceability.

LESSON 7: Product realization

- In the case of implantable medical devices:

1) The records shall include information of components, materials, and conditions for the work environment used, if these could cause the medical device not to meet its specified safety and performance requirements.
2) The suppliers of distribution services or distributors shall maintain records to allow traceability of the products, including the name and address of the shipping package consignee.
3) Records must be maintained and ready for inspection.

LESSON 7: Product realization

- Customer property, provided for use or incorporation into the product, shall be identified, verified, protected, and safeguarded while it is under the organization's control or being used by the organization.
- The organization shall retain records regarding communication with the customer if any customer property is lost, damaged, or found to be unsuitable for use.

LESSON 7: Product realization

- The organization shall establish procedures to preserve product conformity to requirements during its processing, storage, handling and distribution.
- The product shall be protected against alteration, contamination, or damage when exposed to expected conditions and hazards involved in those activities, by documenting and implementing:
  a) Suitable packaging and shipping containers.
  b) Requirements for special conditions, including controls and records, if packaging alone is not sufficient.

LESSON 7: Product realization

- The organization shall determine the necessary monitoring and measuring activities and related equipment, required to provide evidence that products meet their requirements and specifications.
- Documented procedures shall be established to describe the methods and activities to ensure that monitoring and measuring activities are performed according to the requirements.

LESSON 7: Product realization

- To ensure valid results, the monitoring and measuring equipment shall be calibrated or verified prior to its initial use, at specified intervals, against measurement standards traceable to international or national standards.
- If measurement standards do not exist or are not applicable, the calibration or verification method shall be defined, justified and recorded.
- The calibration or verification process of monitoring and measuring equipment shall be documented and records of those activities shall be maintained.

LESSON 7: Product realization

- The monitoring and measuring equipment shall be:
  a) identified to determine the calibration status and validity.
  b) protected from adjustments that would invalidate the measurement results.
  c) protected from damage and deterioration during handling, maintenance and storage.

LESSON 7: Product realization

- If monitoring and measuring equipment is found not to conform to requirements, the organization shall assess and record the validity of the previous measuring results, and take appropriate actions regarding the equipment and products affected.

Refer to standard ISO 10012 for additional information regarding Measurement management systems (Requirements for measurement processes and measuring equipment).

ASSIGNMENT 5

- **Lesson 7:** Product realization

LESSON 8: Measurement, analysis and improvement

- The organization shall plan and implement processes for monitoring, measurement, analysis and improvement, required to:
  a) Demonstrate conformity of product against requirements.
  b) Ensure conformity of the quality management system.
  c) Maintain the effectiveness of the quality management system.
- These processes shall include determination of appropriate methods, including statistical techniques, and the extent of their use.

LESSON 8: Measurement, analysis and improvement

- The organization shall document a process for gathering and monitoring relevant information, to confirm the QMS is effective at meeting customer requirements.
- A feedback process shall be established, based on data from production and post-production activities.
- The feedback information shall be used as input to the risk management process, to ensure conformity with product requirements and continuous improvement of QMS processes.

LESSON 8: Measurement, analysis and improvement

- If specified by the relevant regulatory requirements, the experience obtained from post-production activities shall be used as part of the feedback review process.
- The organization shall establish and maintain documented procedures for handling complaints in a timely manner, according to applicable regulatory requirements.

LESSON 8: Measurement, analysis and improvement

- The complaints handling process shall define the requirements and responsibilities for receiving and recording, evaluating and investigating complaints.
- The process shall define the method of reporting complaints to the appropriate regulatory authorities, handling complaint-related product and how to initiate corrections or corrective actions.
- Any correction or corrective action resulting from the complaint-handling process shall be documented.

LESSON 8: Measurement, analysis and improvement

- When a complaint is not investigated, a justification shall be documented.
- If an investigation of the complaint determines that activities outside the organization contributed to the complaint, the relevant information shall be exchanged between the organization and the external party involved.
- Records of the complaint handling process shall be maintained.

LESSON 8: Measurement, analysis and improvement

- The organization shall identify the applicable regulatory requirements regarding the notification of complaints, and follow a documented procedure when reporting to the appropriate regulatory authorities.
- Records of reporting to those authorities shall be maintained.
- Internal audits shall be conducted by the organization at planned intervals, to confirm the QMS conforms to the requirements of the ISO 13485 standard and other applicable regulatory requirements, and the QMS is effectively implemented and maintained.

LESSON 8: Measurement, analysis and improvement

- The organization shall establish a documented procedure to describe:

1) The responsibilities and requirements for planning and conducting audits.

2) The recording and reporting of audit results.

- The audit program shall be planned, considering the following:

  a) The status and importance of the processes and areas to be audited.

  b) The results of previous audits.

  c) The audit criteria, scope, interval and audit methods.

LESSON 8: Measurement, analysis and improvement

- The organization shall select auditors and conduct the audit to ensure objectivity and the impartiality of the audit process.
- Auditors shall not audit their own work.
- Records shall be maintained, as evidence of the implementation of the audit program and the audit results, including the identification of the processes and areas audited, findings and conclusions.

LESSON 8: Measurement, analysis and improvement

- The management responsible for the audited area shall ensure that nonconformities are addressed, including the implementation of corrections and corrective actions, without unnecessary delays.
- Follow-up activities shall include the verification of the actions taken and the reporting of verification results.

Refer to standard **ISO 19011** for additional information and guidelines for auditing management systems.

LESSON 8: Measurement, analysis and improvement

- The organization shall establish and implement suitable methods for monitoring and measuring processes of the QMS, to confirm the ability of the processes to achieve planned results.
- When planned results are not achieved, corrections and corrective actions shall be implemented, as appropriate.
- The organization shall measure and monitor the characteristics of the product, throughout the different realization processes and stages, to ensure that product requirements have been met.

LESSON 8: Measurement, analysis and improvement

- Monitoring and measuring activities shall be carried out in accordance with the planned and documented arrangements and procedures.
- Records to demonstrate the product's conformity to the specified requirements and acceptance criteria shall be maintained.
- The records shall include information of the test equipment used to perform measurement activities, as appropriate.

LESSON 8: Measurement, analysis and improvement

- Product release and service delivery shall not proceed until all the planned and documented activities have been successfully completed.
- For implantable medical devices, the organization shall record the identity of personnel performing product inspection or testing activities.
- The organization shall ensure that products that do not conform to product requirements are identified and controlled, to prevent the unintended use or delivery of the product.

LESSON 8: Measurement, analysis and improvement

- A procedure shall be established for nonconforming products, to define the controls and related responsibilities and authorities for their identification, documentation, segregation, evaluation and disposition.
- The evaluation of the nonconformity shall determine the need for an investigation and notification to any external party responsible for the issue.
- Records shall be maintained, to capture the nonconformities and any action taken, such as evaluation, investigation, and the rationale for decisions (e.g., product disposition).

LESSON 8: Measurement, analysis and improvement

- When nonconforming product is detected prior to delivery, the organization shall take the required actions to:
  a) Eliminate the detected nonconformity.
  b) Prevent the original intended use or application of the product.
  c) If applicable, authorize the use, release, or acceptance of the product under concession, if justification is provided and approved, and applicable regulatory requirements are met.
- Records of the concession shall be maintained, including the identification of the person authorizing it.

LESSON 8: Measurement, analysis and improvement

- When nonconforming product is detected after delivery or use has started, the organization shall take the required actions, based on the effects or potential effects of the nonconformity.
- Documented procedures shall be followed when issuing advisory notices, as mandated by the applicable regulatory requirements. Those actions shall be ready for implementation at any time.
- Records of the actions taken shall be maintained, also when issuing advisory notices.

LESSON 8: Measurement, analysis and improvement

- The organization shall perform rework following documented procedures, considering the potential adverse effects of the rework on the product.
- The rework procedures shall be reviewed and approved just as the original procedures.
- The reworked product shall be verified to ensure that it meets applicable acceptance criteria and regulatory requirements.
- Records of rework shall be maintained.

LESSON 8: Measurement, analysis and improvement

- Documented procedures shall be established and maintained, to determine, collect and analyze data necessary to demonstrate the suitability, adequacy, and effectiveness of the QMS.
- The procedures shall include determination of appropriate methods and statistical techniques and the extent of their use.

LESSON 8: Measurement, analysis and improvement

- The analysis shall include data gathered from various monitoring and measuring activities, such as feedback, conformity to product requirements, characteristics and trends of processes and products, opportunities for improvement, suppliers data, audits data and service reports.
- The result of the data analysis shall be used as input to improvement activities, when it shows that the quality management system is not suitable, adequate, or effective.
- The records of the data analysis shall be maintained.

LESSON 8: Measurement, analysis and improvement

- The organization shall identify and implement the changes needed for the continued suitability, adequacy, and effectiveness of the QMS, and to maintain the medical device safety and performance.
- The changes shall be implemented using quality processes such as quality policy, quality objectives, audits, post-market surveillance, data analysis, corrective and preventive actions (CAPA process), and management review.

LESSON 8: Measurement, analysis and improvement

- The organization shall implement actions to eliminate the cause of any nonconformity and prevent its reoccurrence, without undue delay.
- Those actions shall be proportionate to the effects of the nonconformity.
- A procedure shall be established to define requirements for:
  a) Reviewing nonconformities and complaints.
  b) Determining the root causes of nonconformities.
  c) Evaluating the need for action to prevent reoccurrence.

LESSON 8: Measurement, analysis and improvement

d) Planning, documenting and implementing actions needed, including documentation updates, as required.

e) Verifying that the corrective action does not adversely affect the ability to meet applicable regulatory requirements or the safety and performance of the medical device.

f) Reviewing the effectiveness of the corrective actions taken.

- Records of the corrective actions, including investigation results and actions taken shall be maintained.

LESSON 8: Measurement, analysis and improvement

- The organization shall implement actions to eliminate the causes of potential nonconformities and prevent they occurrence.
- Those actions shall be proportionate to the effects of the potential nonconformities.
- A procedure shall be established to define requirements for:
  a) Determining potential nonconformities and their causes.
  b) Evaluating the need for action to prevent occurrence.

LESSON 8: Measurement, analysis and improvement

c) Planning, documenting and implementing actions needed, including documentation updates, as required.

d) Verifying that the preventive action does not adversely affect the ability to meet applicable regulatory requirements or the safety and performance of the medical device.

e) Reviewing the effectiveness of the preventive actions taken.

- Records of the preventive actions, including investigation results and actions taken shall be maintained.

ASSIGNMENT 6

- **Lesson 8:** Measurement, analysis and improvement

FINAL ASSIGNMENT

- **Lesson 1:** Why do we need a Quality Management System?
- **Lesson 2:** Scope and applicability of ISO 13485:2016
- **Lesson 3:** General QMS requirements
- **Lesson 4:** Documentation requirements
- **Lesson 5:** Management responsibility
- **Lesson 6:** Resource management
- **Lesson 7:** Product realization
- **Lesson 8:** Measurement, analysis and improvement

CONGRATULATIONS!

ISO 13485:2016

Medical Devices

Quality management systems

Requirements for regulatory purposes

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